

Course Name: Advance C-Programming

Course Code: **Internal Marks: 25** **External Marks: 75** **Credit: 4**

UNIT-I

Arrays:- Definition, declaration and initialization of one dimensional array; Accessing array elements; Displaying array elements; Sorting arrays; Arrays and function. Two-Dimensional array: Declaration and Initialization, Accessing and Displaying, Memory representation of array [Row Major, Column Major]; Multidimensional array.

UNIT-II

Pointers:- Definition and declaration, Initialization; Indirection operator, address of operator; pointer arithmetic; dynamic memory allocation; arrays and pointers; function and pointers.

UNIT-III

Strings:- Definition, declaration and initialization of strings; standard library function: strlen(), strcpy(), strcat(), strcmp(); Implementation without using standard library functions.

UNIT-IV

Structures:- Definition and declaration; Variables initialization; Accessing fields and structure operations; Nested structures; Union: Definition and declaration; Differentiate between Union and structure.

UNIT-V

Introduction C Preprocessor:- Definition of Preprocessor; Macro substitution directives; File inclusion directives; Conditional compilation. Bitwise operators; Shift operators; Masks; Bit field.

UNIT-VI

File Handling:- Definition of Files, Opening modes of files; Standard function: fopen(), fclose(), feof(), fseek(), fwind(); Using text files: fgetc(), fputc(), fscanf(), Command line arguments.